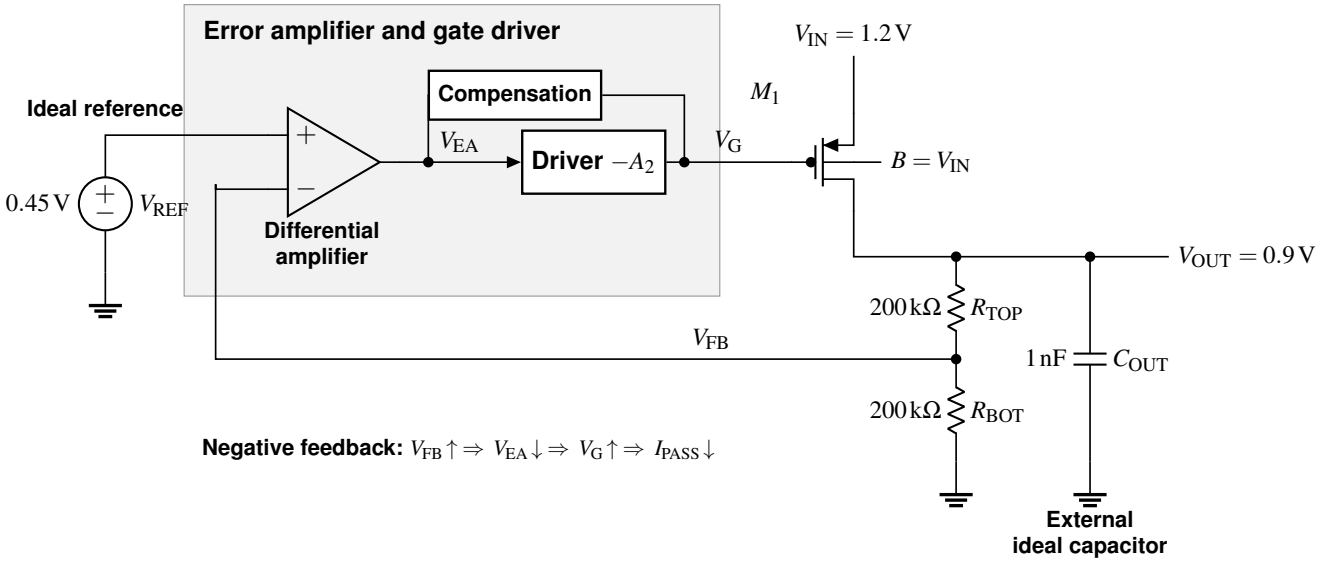
1. LDO regulation architecture

130 nm CMOS power delivery — historical published circuits



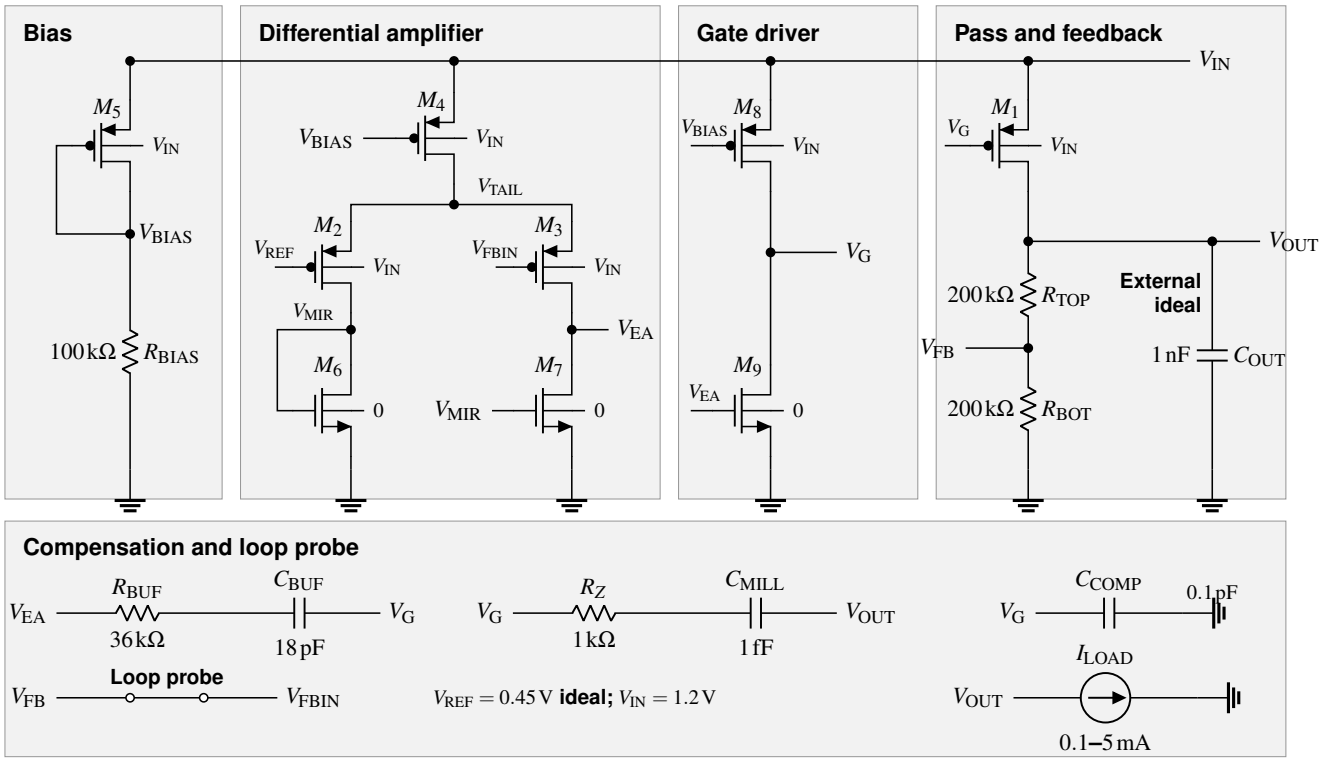
1. LDO regulation architecture. Historical r6. The reference and external output capacitor are ideal testbench elements. The amplifier, gate driver and PMOS pass stage use transistor models. The compensation network is shown functionally; its complete topology is on the following page.

Source: `report/figures/ldo-architecture.tex`. The PDF and SVG share this drawing source.

Schematic-level circuit documentation. See the project device inventories and result definitions for complete conditions.

2. LDO transistor circuit

130 nm CMOS power delivery — historical published circuits



Device	M_1	M_2, M_3	M_4, M_8	M_5	M_6, M_7	M_9
W_{unit}/L (µm)	10/0.15	2.5/1	2.5/1	2.5/1	2/1	2/1
m; $n_f = 1$ for all	**10**	**4**	**8**	**2**	**2**	**4**

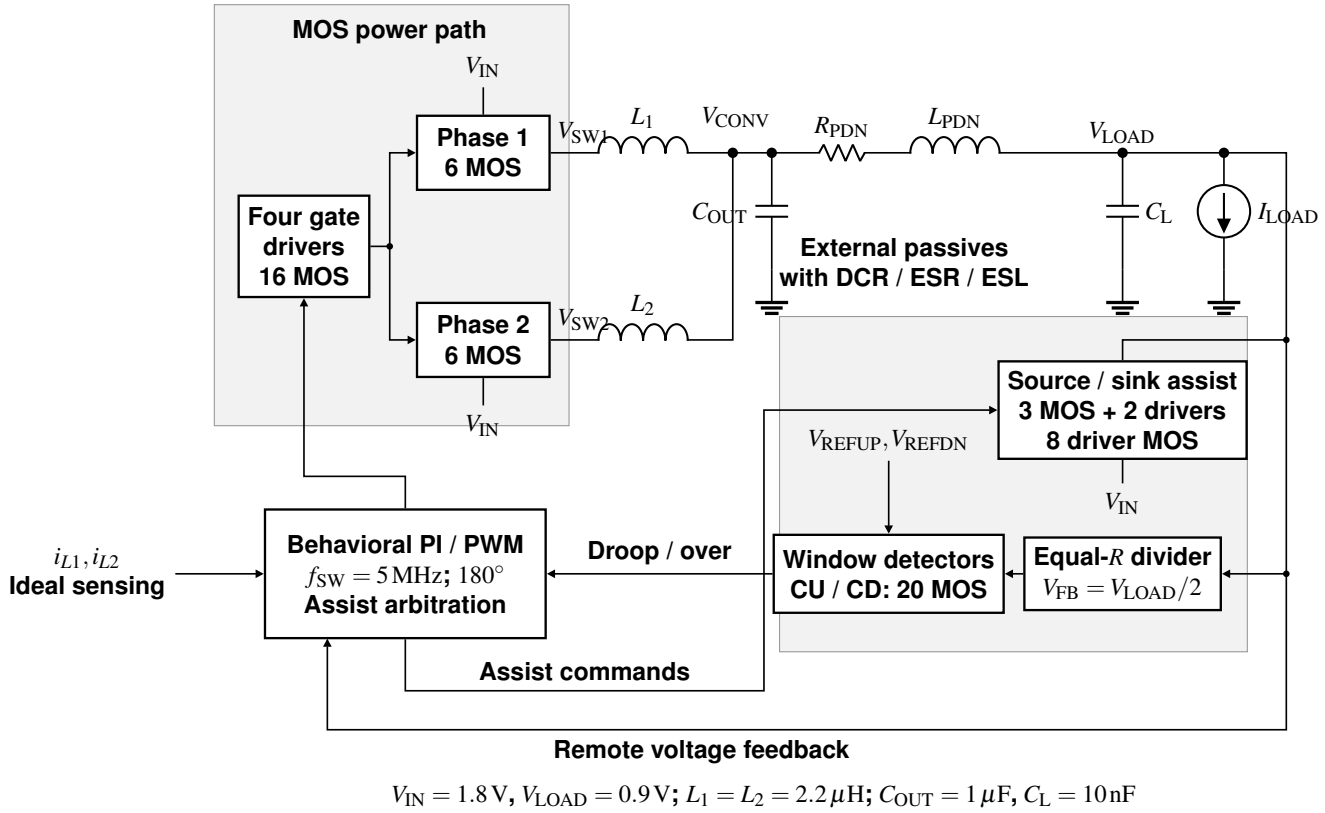
2. LDO transistor circuit. All nine MOS instances of r6. Matching net labels denote electrical connections. Filled dots identify junctions; unmarked crossings are not connected. Dimensions are in micrometers, m is parallel multiplicity and $n_f = 1$. Body connections are labeled separately. In the source inventory, V_{EA} maps to EAOUT, V_G to GATE, and V_{MIR} to MIRROR; ground is node 0.

Source: `report/figures/ldo-core.tex`. The PDF and SVG share this drawing source.

Schematic-level circuit documentation. See the project device inventories and result definitions for complete conditions.

3. Buck system architecture

130 nm CMOS power delivery — historical published circuits



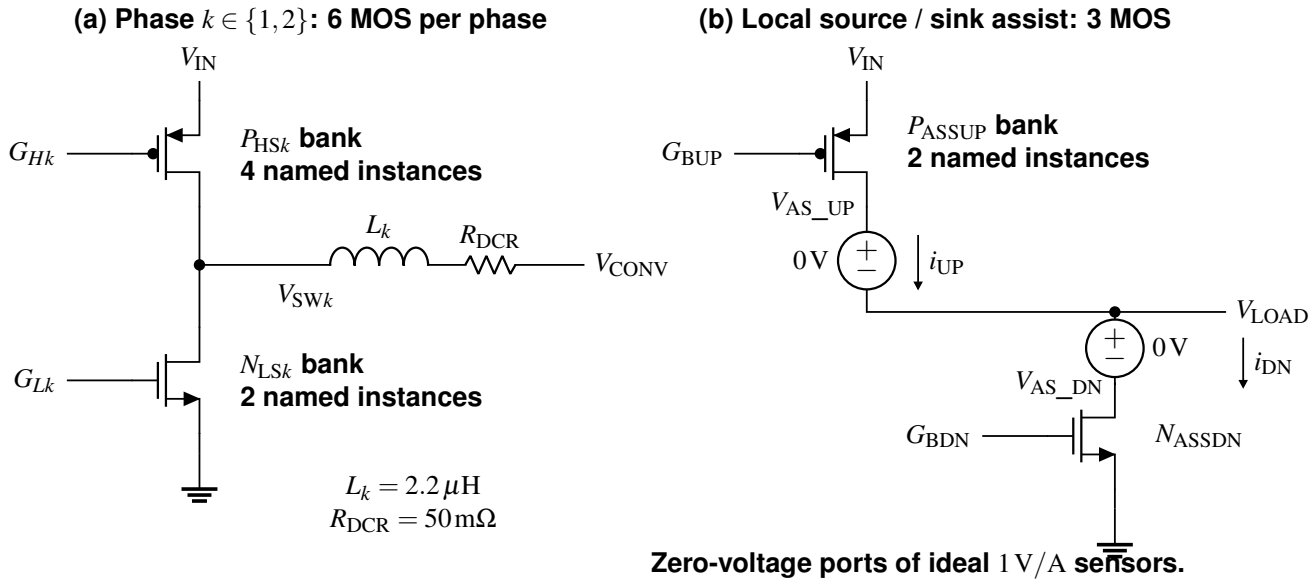
3. Buck system architecture. Two interleaved phases regulate the remote load. Local MOS assist and coordinated PI-state handoff both contribute to the ON condition. PI/PWM control and current sensing are behavioral; the PDN and external passives are parameterized test conditions.

Source: `report/figures/buck-architecture.tex`. The PDF and SVG share this drawing source.

Schematic-level circuit documentation. See the project device inventories and result definitions for complete conditions.

4. Buck power stage and assist

130 nm CMOS power delivery — historical published circuits



Bank / instance membership	W_{unit}/L (μm)	m	n_f	W_{bank} (μm)
P_{HSk}: PHSk, PHSk_B1, _B2, _B3	10/0.5	100	1	4000
N_{LSk}: NLSk, NLSk_B1	10/0.5	100	1	2000
P_{ASSUP}: PASSUP, PASSUP_B1	10/0.5	60	1	1200
N_{ASSDN}: PASSDN	10/0.5	60	1	600

PMOS bulks: V_{IN}; NMOS bulks: ground.

r9: $G_{BUP,DRV} \xrightarrow{1\,k\Omega} G_{BUP}$, $G_{BDN,DRV} \xrightarrow{2\,k\Omega} G_{BDN}$; unchanged MOS dimensions.

4. Buck power stage and assist. Parallel-bank labels enumerate the power devices. The r9 variation retains all MOS dimensions and inserts the stated assist gate resistances. Identical net labels are connected.

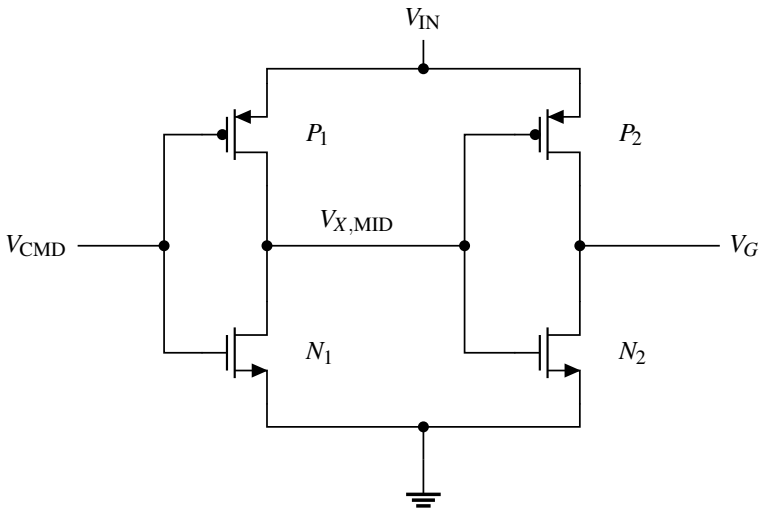
Source: `report/figures/buck-power.tex`. The PDF and SVG share this drawing source.

Schematic-level circuit documentation. See the project device inventories and result definitions for complete conditions.

5. Gate-driver circuits

130 nm CMOS power delivery — historical published circuits

Driver template X: two cascaded inverters



Instance suffix	W_{unit}/L (μm)	m	n_f	W_{total} (μm)
_P1	$8/0.5$	2	1	16
_N1	$4/0.5$	2	1	8
_P2	$20/0.5$	20	1	400
_N2	$10/0.5$	20	1	200

Prefix X	V_{CMD}	$V_{X,MID}$	V_G (r8)
DH1	**CMDH1**	**DH1_MID**	**GH1**
DH2	**CMDH2**	**DH2_MID**	**GH2**
DL1	**CMDL1**	**DL1_MID**	**GL1**
DL2	**CMDL2**	**DL2_MID**	**GL2**
DAU	**CMDUP**	**DAU_MID**	**GBUP**
DAD	**CMDDN**	**DAD_MID**	**GBDN**

Six templates $\times$ four named MOS = 24 MOS.
PMOS bulks: V_{IN}; NMOS bulks: ground.
r9: DAU / DAD outputs become GBUP_DRV / GBDN_DRV;
$1\,k\Omega$ / $2\,k\Omega$ connect them to GBUP / GBDN.

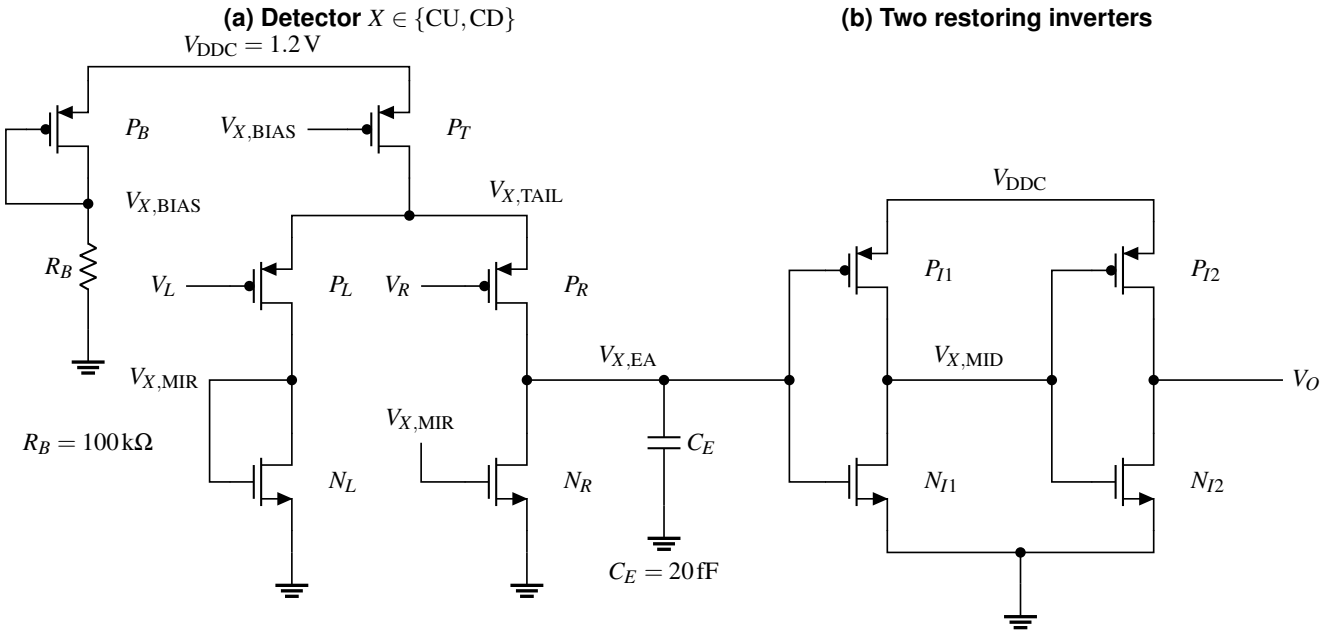
5. Gate-driver circuits. Six instances use the common two-stage inverter topology and the displayed instance/terminal mapping. Each instance contains four MOS devices; two inversions preserve command polarity.

Source: `report/figures/buck-driver.tex`. The PDF and SVG share this drawing source.

Schematic-level circuit documentation. See the project device inventories and result definitions for complete conditions.

6. Window-detector circuits

130 nm CMOS power delivery — historical published circuits



Suffix	W_{unit}/L (μm)	m	n_f	W_{total} (μm)
_PB	2.5/1	1	1	2.5
_PT	2.5/1	16	1	40
_PL, _PR	5/0.5	4	1	20 **each**
_NL, _NR	2/1	4	1	8 **each**
_PI1, _PI2	2/0.15	1	1	2 **each**
_NI1, _NI2	1/0.15	1	1	1 **each**

X	V_L	V_R	V_O
CU	**REFUP**	**FB**	**DROOP_RAW**
CD	**FB**	**REFDN**	**OVER_RAW**

$$V_{\mathrm{FB}} = V_{\mathrm{LOAD}}/2$$
$$V_{\mathrm{REFUP}} = (V_{\mathrm{REQ}} - V_{\mathrm{WINDOW}})/2$$
$$V_{\mathrm{REFDN}} = (V_{\mathrm{REQ}} + V_{\mathrm{WINDOW}})/2$$

Two templates $\times$ ten named MOS = 20 MOS. PMOS bulks: V_{DDC}; NMOS bulks: ground.
Equal divider resistors: $200\,\mathrm{k\Omega}$**. Ideal threshold references:** $0.4375/0.4625\,\mathrm{V}$**.**
$V_{\mathrm{REQ}} = 0.9\,\mathrm{V}$**,** $V_{\mathrm{WINDOW}} = 25\,\mathrm{mV}$**; nominal output window:** 0.875**–**$0.925\,\mathrm{V}$**.**

6. Window-detector circuits. Two ten-MOS circuits use the template and mapping shown. Input assignment distinguishes low and high voltage detection. Matching net labels denote electrical connection. Reference voltages are bench conditions.

Source: `report/figures/buck-detector.tex`. The PDF and SVG share this drawing source.

Schematic-level circuit documentation. See the project device inventories and result definitions for complete conditions.